ECA08 - ANALOG AND DIGITAL COMMUNICATION

CO2 CASE STUDY ASSIGNMENT REPORT

FM Broadcast System: From Angle-Modulation Selection to Noise-Performance Deployment

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Reg No: 192512355	**Bloom's Taxonomy Levels:** K4 (Analyze), K5 (Evaluate), K6 (Design)

1. Design Parameter Specifications Baseline

Design Parameter	**Specification / Constraint Value**
Information Signal	Broadcast Audio, maximum message frequency f_m = 15 kHz
Candidate Carrier Frequency	100 MHz (VHF FM Broadcast Band)
Peak Frequency Deviation (Δf)	75 kHz for standard wideband FM design
Channel Condition	AWGN plus practical receiver limitations and adjacent-channel interference
Receiver Architecture	Superheterodyne FM Broadcast Receiver with Limiter and Frequency Discriminator
Noise Improvement Scheme	Pre-emphasis (75 μs) at Transmitter & De-emphasis (75 μs) at Receiver
Reference Systems	AM (DSB-FC), DSB-SC, FM, and PM
Analysis & Simulation Platform	MATLAB

STAGE 1: Interpret the Broadcast System Requirement

1.1 Basic Communication-System Architecture & Modulation Need

The communication chain operates through standard functional blocks: Information Source → Pre-emphasis/Low-Pass Filter → Modulator → Power Amplifier → AWGN Channel → RF/IF Limiter-Discriminator Receiver → De-emphasis Filter → Loudspeaker.

- **Antenna Size Constraints:** Direct baseband audio (15 kHz) cannot be efficiently radiated as electromagnetic waves since it requires an impractically large antenna ($L = \lambda/2 = 10$ km). Modulating onto a 100 MHz carrier reduces antenna size to 1.5 meters.
- **Frequency Allocation:** Modulation enables Frequency Division Multiplexing (FDM), allowing multiple stations to broadcast simultaneously in the VHF band without spectral overlap.

1.2 Angle Modulation & Constant-Average-Power Property

Angle modulation preserves total transmitted envelope power constant regardless of the modulating signal amplitude. In ideal FM/PM, the envelope A_c is constant:

Constant Power Property: $P_{avg} = (A_c^2 / 2R)$. Unlike AM, power is redistributed among infinite Bessel sidebands without increasing total transmitted power during high modulation depths.

1.3 Major Noise Sources & Channel Conditions

- **Additive White Gaussian Noise (AWGN):** Thermal noise generated in receiver front-end conductors and semiconductor devices, modeled as zero-mean Gaussian distributed with flat power spectral density $N_0/2$.
- **Interference Sources:** Man-made RF interference, atmospheric noise, and adjacent-channel spectral spill from nearby stations.

CASE STUDY DECISION 1: BROADCAST SYSTEM REQUIREMENT STATEMENT

I recommend a Wideband FM operating basis centered at $f_c = 1000$ MHz with $\Delta f = 75$ kHz, audio cutoff $f_m = 15$ kHz, pre-emphasis/de-emphasis time constant $\tau = 75$ μs, and a minimum operational C/N of 12 dB.

Justification: Satisfies high-fidelity broadcast standards while ensuring complete noise immunity above the FM threshold within standard 200 kHz channel allocations.

STAGE 2: Investigate and Select the Angle-Modulation Scheme

2.1 Mathematical Expressions

- **Frequency Modulation (FM):** $s_{FM}(t) = A_c \cos(2\pi f_c t + 2\pi k_f \int m(\tau)\, d\tau)$. Instantaneous frequency $f_i(t) = f_c + k_f m(t)$. Modulation index $\beta = \Delta f / f_m = 75$ kHz / 15 kHz = 5.
- **Phase Modulation (PM):** $s_{PM}(t) = A_c \cos(2\pi f_c t + k_p m(t))$. Instantaneous frequency $f_i(t) = f_c + (k_p / 2\pi)(dm/dt)$. Modulation index $\beta_p = k_p \max|m(t)|$.

2.2 Quantitative Comparison: FM vs. PM

Parameter	**Frequency Modulation (FM)**	**Phase Modulation (PM)**	**Impact Analysis**
Frequency Deviation (Δf)	Independent of audio frequency ($\Delta f = k_f A_m$)	Proportional to audio frequency ($\Delta f = k_p A_m f_m$)	FM avoids high-frequency bandwidth explosion
Modulation Index (β)	$\beta = \Delta f / f_m$ (Decreases with higher audio frequency)	$\beta_p = k_p A_m$ (Constant over audio frequency)	FM provides uniform SNR boost across baseband
Noise Performance	Parabolic noise density; easily suppressed by De-emphasis	Flat noise density; inferior high-frequency SNR boost	FM provides superior audio SNR
Receiver Complexity	Limiter + Frequency Discriminator / PLL	Coherent Phase Demodulator / Complex carrier recovery	FM receiver design is much simpler

CASE STUDY DECISION 2: ANGLE-MODULATION SCHEME SELECTION
I recommend Frequency Modulation (FM) over Phase Modulation (PM). Justification: In PM, frequency deviation scales linearly with message frequency ($\Delta f = k_p$

A_m f_m), causing severe bandwidth explosion at high audio frequencies (15 kHz). FM maintains fixed peak frequency deviation (Δf = 75 kHz), optimizing bandwidth utilization while enabling simple discriminator detection.

STAGE 3: Analyze FM Spectrum and Transmission Bandwidth

3.1 Bessel-Function Sideband Analysis & Carson's Rule

For single-tone sinusoidal FM with β = 75 kHz / 15 kHz = 5, the spectrum consists of a carrier and symmetrical sidebands spaced at ± n f_m:

- **Bessel Coefficients (β = 5):** $J_0(5) = -0.178$, $J_1(5) = -0.328$, $J_2(5) = 0.047$, $J_3(5) = 0.365$, $J_4(5) = 0.391$, $J_5(5) = 0.261$, $J_6(5) = 0.131$, $J_7(5) = 0.053$, $J_8(5) = 0.018$.
- **Significant Sideband Bandwidth:** Significant sidebands contain components with $|J_n(\beta)| \geq 0.01$. For β = 5, there are n = β + 1 = 6 to 8 significant sideband pairs. Using n = 8: BW_sig = 2 * n * f_m = 2 * 8 * 15 kHz = 240 kHz.
- **Carson's Rule Bandwidth:** B_FM = 2 (Δf + f_m) = 2 (75 kHz + 15 kHz) = 2 (90 kHz) = 180 kHz.

3.2 Regulatory Framework (ITU-R & FCC 47 CFR §73.322)

The FCC and ITU-R specify a standard 200 kHz channel allocation for commercial FM broadcasting. Carson's rule bandwidth of 180 kHz fits cleanly inside 200 kHz, leaving a 20 kHz guard band (10 kHz on each band edge) to eliminate adjacent-channel interference.

CASE STUDY DECISION 3: FREQUENCY DEVIATION & BANDWIDTH RECOMMENDATION

I recommend Peak Frequency Deviation Δf = 75 kHz resulting in Carson's Bandwidth B_FM = 180 kHz.

Justification: Balances audio fidelity and noise reduction (β = 5) while strictly adhering to the 200 kHz channel spacing mandated by FCC §73.322 and ITU-R BS.450.

STAGE 4: Develop Transmitter and Receiver Architecture

4.1 AM/FM Transmitter & Receiver Architecture

- **FM Transmitter Chain:** Audio Source → Pre-emphasis Filter (75 μs) → Armstrong Indirect FM Modulator (NBFM Generator → Frequency Multiplier Chain) → Class-C Power Amplifier → Antenna.

- **FM Superheterodyne Receiver Chain:** Antenna → RF Amplifier → Mixer (f_{LO} = 110.7 MHz) → IF Amplifier (f_{IF} = 10.7 MHz) → Hard Limiter → Frequency Discriminator / Foster-Seeley / Ratio Detector → De-emphasis Filter (75 μs) → Audio Power Amplifier → Loudspeaker.

4.2 IF Selection & Detection Mechanisms

- **Intermediate Frequency:** f_{IF} = 10.7 MHz is chosen as the standard broadcast IF, eliminating image interference (f_{img} = 122.1 MHz, outside 88-108 MHz band).
- **Hard Limiter:** Removes amplitude variations caused by AWGN noise and fading before discriminator detection.
- **Frequency Discriminator:** Converts frequency variations into proportional voltage envelope variations for baseband audio recovery.

CASE STUDY DECISION 4: ARCHITECTURE SELECTION
I recommend the Armstrong Indirect FM Transmitter coupled with a Superheterodyne Limiter-Discriminator Receiver operating at f_{IF} = 10.7 MHz.

Justification: Indirect FM provides crystal-frequency stability. The 10.7 MHz IF Superheterodyne receiver delivers high sensitivity, sharp adjacent-channel selectivity, and robust AM noise rejection via hard limiting.

STAGE 5: Analyze AM and DSB-SC Noise Performance

5.1 Mathematical Noise Models & SNR Calculation

Under AWGN conditions with baseband noise power $P_n = N_0 f_m$ and carrier power P_c:

- **Standard AM (DSB-FC):** $(SNR)_{o,AM} = (m^2 / (2 + m^2)) * (P_c / (N_0 f_m))$. For m = 1.0, $(SNR)_{o,AM} = (1/3) (SNR)_{channel}$. For m = 0.7, $(SNR)_{o,AM} = 0.1968 (SNR)_{channel}$.
- **DSB-SC:** $(SNR)_{o,DSB} = P_c / (N_0 f_m) = (SNR)_{channel}$. DSB-SC achieves a 4.8 dB noise performance advantage over AM (m = 0.7).

5.2 AM vs. DSB-SC Summary Matrix

System	Output SNR Formula	Power Efficiency	Demodulator Complexity
Standard AM (m = 0.7)	$(SNR)_o = 0.1968 * (SNR)_{channel}$	19.68% (Wastes 80% power)	Very Low (Envelope Detector)

DSB-SC	$(SNR)_o = 1.0 * (SNR)_{channel}$	100% (Sideband power)	High (Coherent / Costas Loop)
Wideband FM ($\beta = 5$)	$(SNR)_o = 3\beta^2 (C/N)_{channel} = 75 * (C/N)$	100% (Constant Envelope)	Moderate (Limiter-Discriminator)

STAGE 6: Evaluate FM Noise, Pre-Emphasis/De-Emphasis & Threshold

6.1 FM Output SNR & Noise Improvement

The output SNR for wideband FM above threshold is given by:

FM SNR Formula: $(SNR)_{o,FM} = 3\beta^2 (B_{FM} / 2 f_m) (C/N) = 3\beta^2 (1 + \beta) (C/N)$

For $\beta = 5$, $(SNR)_{o,FM} = 3(25)(6) (C/N) = 450 (C/N) \equiv (C/N)_{dB} + 26.5$ dB. This demonstrates a massive 26.5 dB SNR enhancement over baseline linear modulation.

6.2 Pre-Emphasis and De-Emphasis Operation

- **Parabolic Noise Problem:** The noise PSD at the FM discriminator output is parabolic ($S_n(f) \propto f^2$), heavily corrupting high audio frequencies.
- **Complementary Solution:** Pre-emphasis boosts high audio frequencies at the transmitter above cutoff frequency $f_1 = 1 / (2\pi \tau) = 2.12$ kHz (for $\tau = 75$ μs). De-emphasis attenuates high frequencies at the receiver, restoring flat audio response while suppressing parabolic high-frequency noise.
- **SNR Gain:** Pre-emphasis/de-emphasis yields an additional SNR improvement of approx. 13 dB for broadcast audio.

6.3 FM Threshold Effect

When input C/N drops below ~10 to 12 dB, sudden noise spikes ('popcorn noise') occur due to phase doublet slips, causing rapid degradation of output SNR. Operating C/N must be maintained ≥ 12 dB.

CASE STUDY DECISION 5: OPERATING CONDITIONS & PRE-EMPHASIS RECOMMENDATION

I recommend adopting 75 μs Pre-emphasis / De-emphasis processing and maintaining an operating Carrier-to-Noise Ratio (C/N) ≥ 13 dB.

Justification: Guarantees operation safely above the 10 dB FM threshold while delivering an overall recovered audio SNR > 45 dB.

STAGE 7: MATLAB Verification & Final Deployment Decision

7.1 Final Specification Verification Matrix

Requirement Parameter	Specified / Theoretical Target	Achieved / Verified Value	Satisfied?
Message Bandwidth (f_m)	15 kHz	15 kHz	**YES**
Carrier Frequency (f_c)	100 MHz	100 MHz	**YES**
Peak Deviation (Δf)	75 kHz	75 kHz	**YES**
FM Modulation Index (β)	Calculated (β = 5)	5.0	**YES**
Transmission Bandwidth	Carson's Rule ≤ 200 kHz	180 kHz (20 kHz Guard)	**YES**
Noise Condition	AWGN Channel	AWGN + Threshold Model	**YES**
Pre-emphasis / De-emphasis	75 µs Time Constant	75 µs (13 dB Noise Gain)	**YES**
Receiver Architecture	Superheterodyne Limiter-Discriminator	10.7 MHz IF Superhet	**YES**
MATLAB Verification	Waveforms, Spectra, Threshold Plots	Verified via Simulation	**YES**

FINAL ENGINEERING DEPLOYMENT DECISION: YES (DEPLOY)
DEPLOYMENT VERDICT: YES

Engineering Defense: The proposed Wideband FM broadcast system ($\Delta f = 75$ kHz, $\beta = 5$, $B_FM = 180$ kHz) operating at 100 MHz carrier frequency satisfies all FCC §73.322 and ITU-R BS.450 broadcast requirements. Combining Armstrong indirect FM transmission, a 10.7 MHz IF Superheterodyne Limiter-Discriminator receiver, 75 μs pre-emphasis/de-emphasis, and maintaining operating $C/N \geq 13$ dB achieves high recovered audio fidelity ($SNR > 45$ dB) with robust noise immunity.